

XVIII Data Academy

DevOps Fundamentals

First Dive

DevOps, Git & Collaboration for Data Engineering

What you will be able to do by the end of this First Dive

1. Understand what is DevOps culture and how its concepts apply to data world
2. Create and structure a simple Git repository for a data project
3. Understand the role of CI/CD in data pipelines
4. Grasp what DataOps means in practice and why it matters for data teams
5. Work with branches and Pull Requests (PRs) in a collaborative workflow

**WHEN THE CODE WORKS
ON MY MACHINE:**



**WHEN THE CODE GOES
TO PRODUCTION:**



**WORKED FINE IN
DEV**

OPS PROBLEM NOW



What is DevOps?

DevOps is not.

**BUSTING
THE
MYTHS**



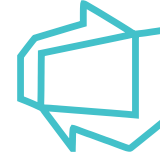
METHODOLOGY

that you have to follow rigidly to achieve something.



SOFTWARE

specific piece of software or software bundle that you buy, install and are ready to use.



AZURE DEVOPS

is a platform that provides an end-to-end DevOps toolchain for developing and deploying software.



AGILE

is a different approaches but combined lead to great efficiencies and more reliable results. DevOps seeks to improve processes, while Agility focuses on adaptability to answer the needs of customers.



TEAM OR JOB TITLE

that make DevOps happen (although these titles are reasonably common in the industry).



QUICK FIX

for an organization's challenges and when overcome / resolved it is abandoned.



PREDECESSOR OF THE SRE

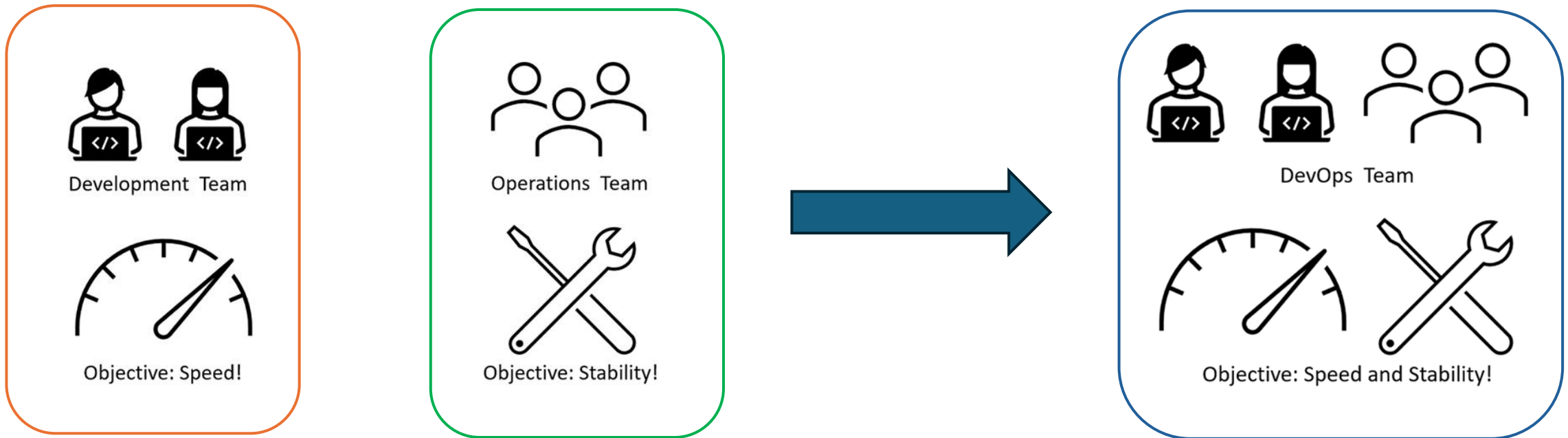
They complement each other. Site Reliability Engineering culture prioritizes reliability, while DevOps accentuates agility across all stages of the SDLC.



CI/CD

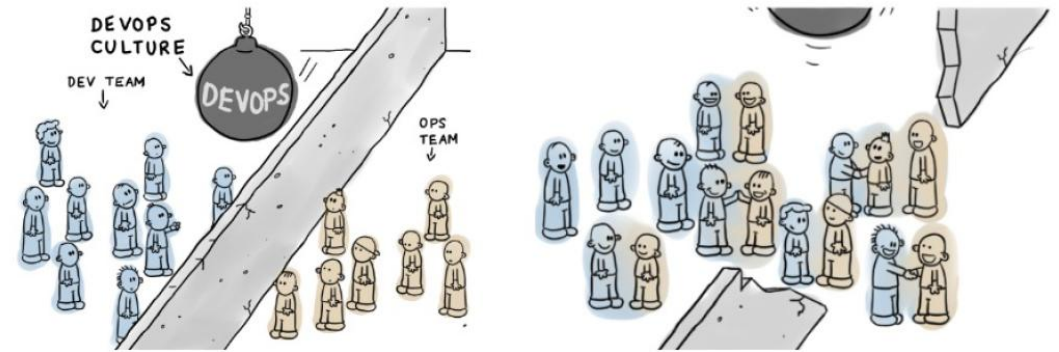
Is one of the practices of DevOps, which makes use of the tools to implement agile development and automate deliver code changes more frequently and reliably (with quality).

Development vs. Operation



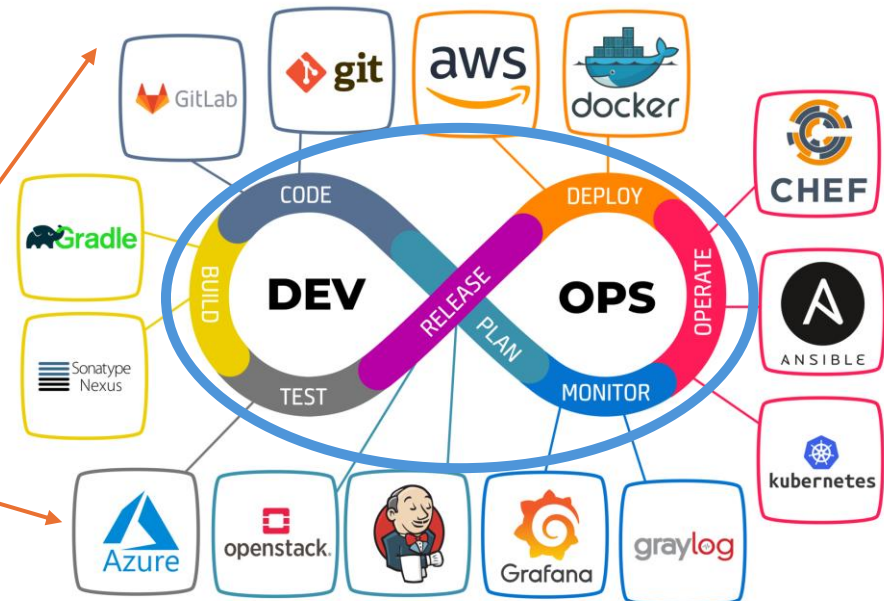
That's the problem that we hope to resolve is to get these two teams rowing in the same direction.

That's the goal of the DevOps cultural shift, is to bring these two teams together and give them a common set of goals that they are cooperatively working towards.



DevOps is the union of **people**, **process**, and **products** to enable continuous delivery of value to our end users.

Donovan Brown (ex-Principal DevOps Manager of Microsoft)



Culture

around thinking, problem solving, automation, delivering high quality results, teamwork, and execution.

people,

must be involved, share learning, and be trained to help in this transformation.



Before choosing tools, we must focus on the **practices** how teams work and share responsibilities.

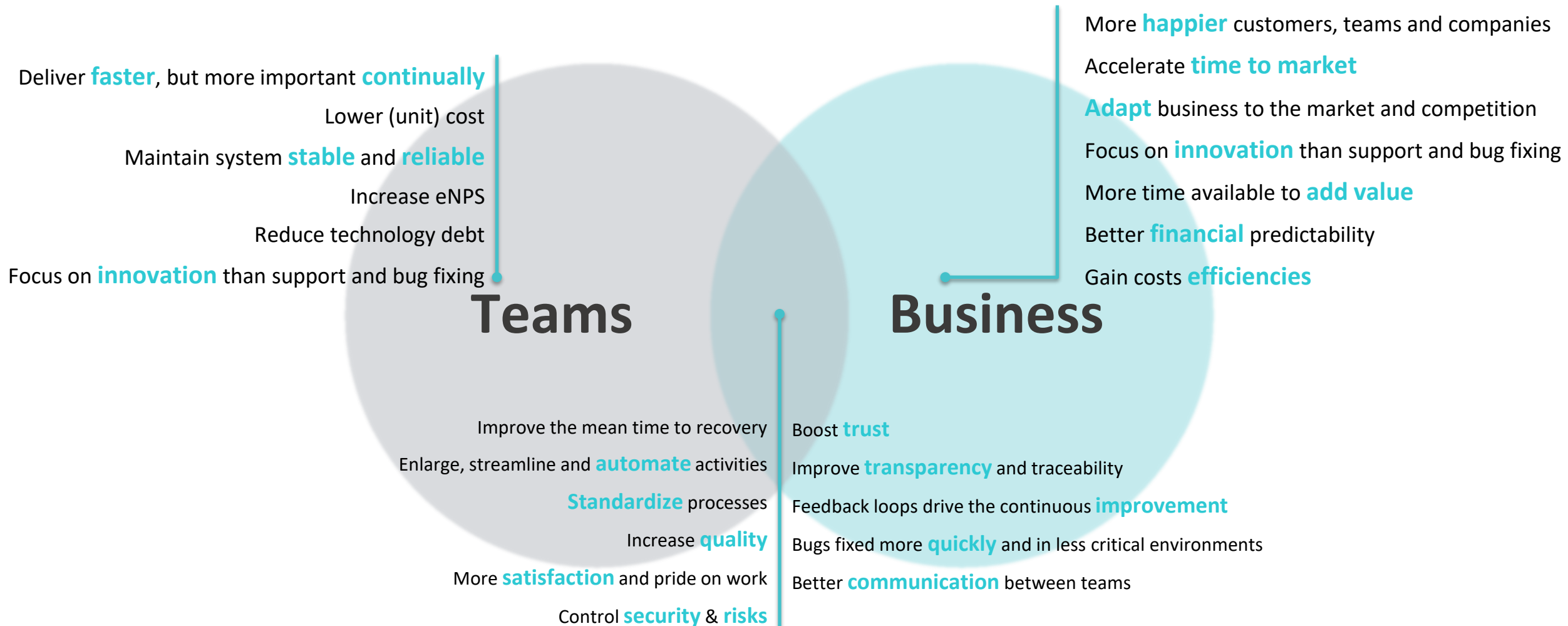


There are "tons" of **tools** and **technologies** on the market. They are important, but they are a way to **facilitate** our journey.

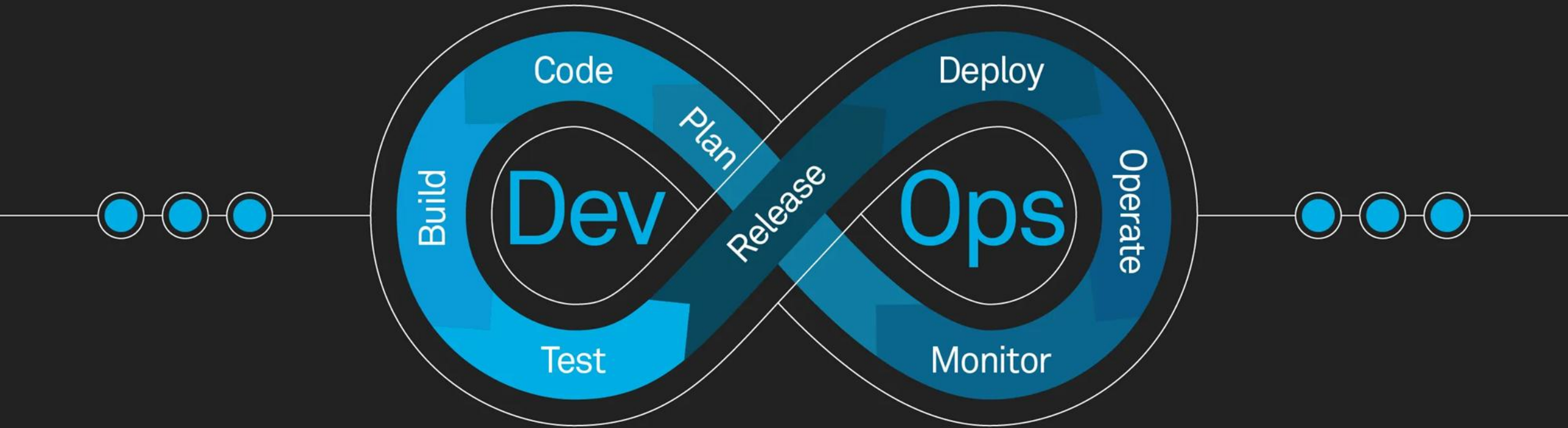


What are the Benefits?

Teams and Companies that adopt DevOps culture, practices, and tools can see advantages at some levels:



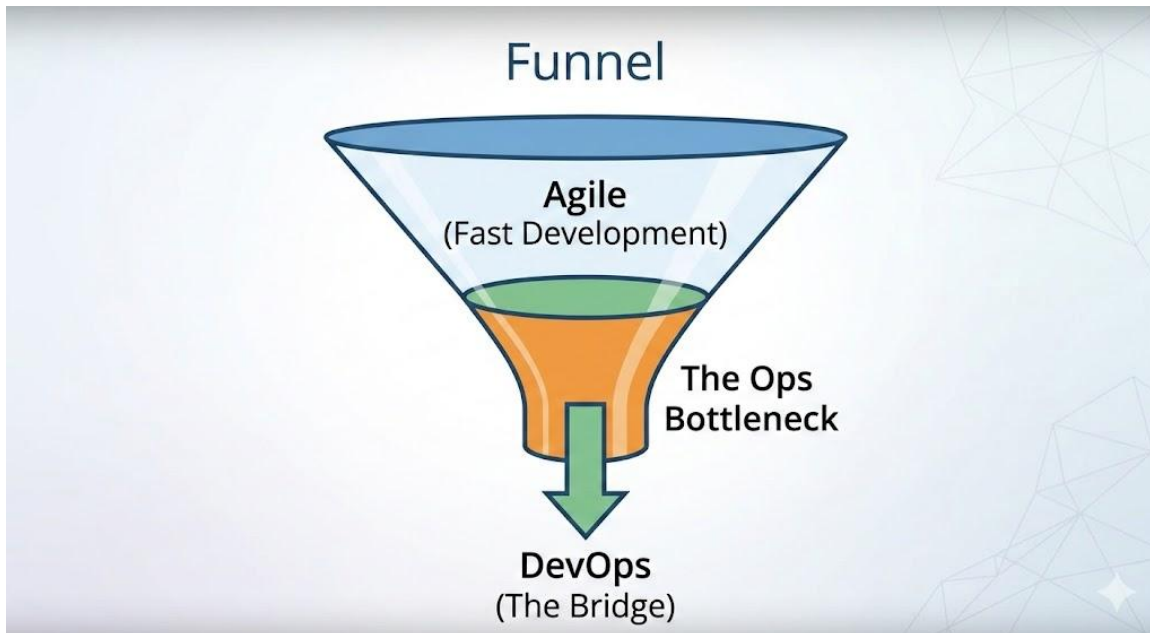
DevOps Loop





What is Agile?

Agile & DevOps: The Perfect Synergy



- 🏃 **The Agile Revolution (Early 2000s)**
 - Smaller batches, faster Dev cycles.
- 📄 **The Agile Manifesto (Core Values)**
 - *Individuals & Interactions* (over processes)
 - *Working Software* (over documentation)
 - *Customer Collaboration* (over contracts)
 - *Responding to Change* (over following a plan)
- ⚠️ **The Unintended Bottleneck**
 - Agile speed up Dev, but excluded Ops.
- 🌉 **The Birth of DevOps**
 - A grassroots movement to close the Dev → Ops gap.

Agile & DevOps: The Perfect Synergy

Manifesto for Agile Software Development

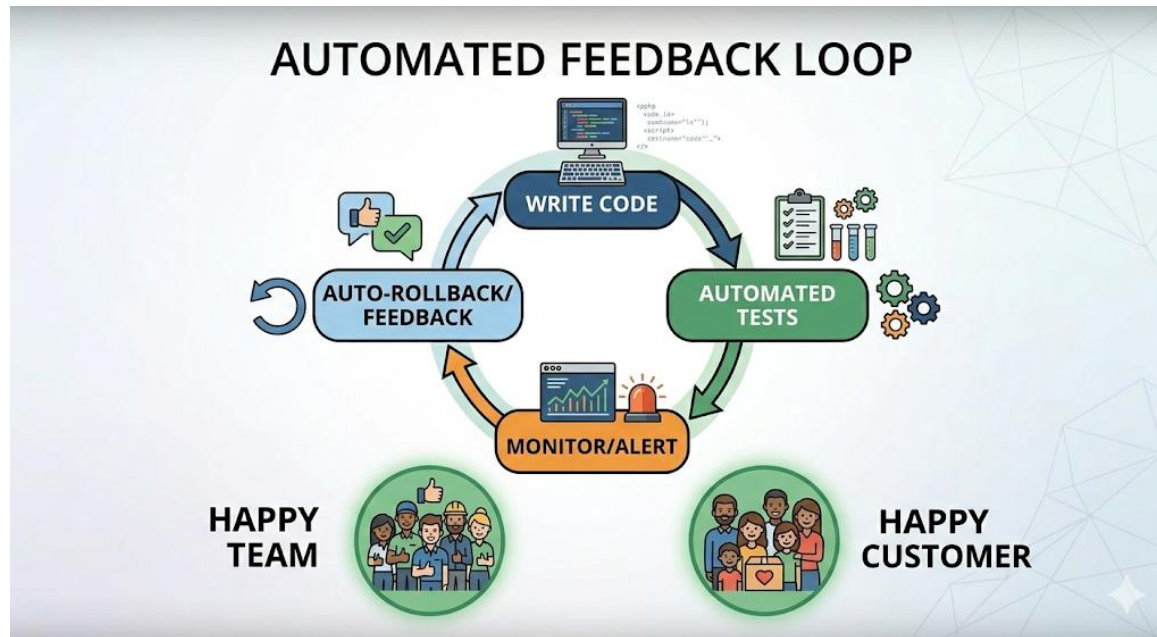
We are uncovering better ways of developing software by doing it and helping others do it.
Through this work we have come to value:

Individuals and interactions over processes and tools
Working software over comprehensive documentation
Customer collaboration over contract negotiation
Responding to change over following a plan

That is, while there is value in the items on the right, we value the items on the left more.

Kent Beck	James Grenning	Robert C. Martin
Mike Beedle	Jim Highsmith	Steve Mellor
Arie van Bennekum	Andrew Hunt	Ken Schwaber
Alistair Cockburn	Ron Jeffries	Jeff Sutherland
Ward Cunningham	Jon Kern	Dave Thomas
Martin Fowler	Brian Marick	

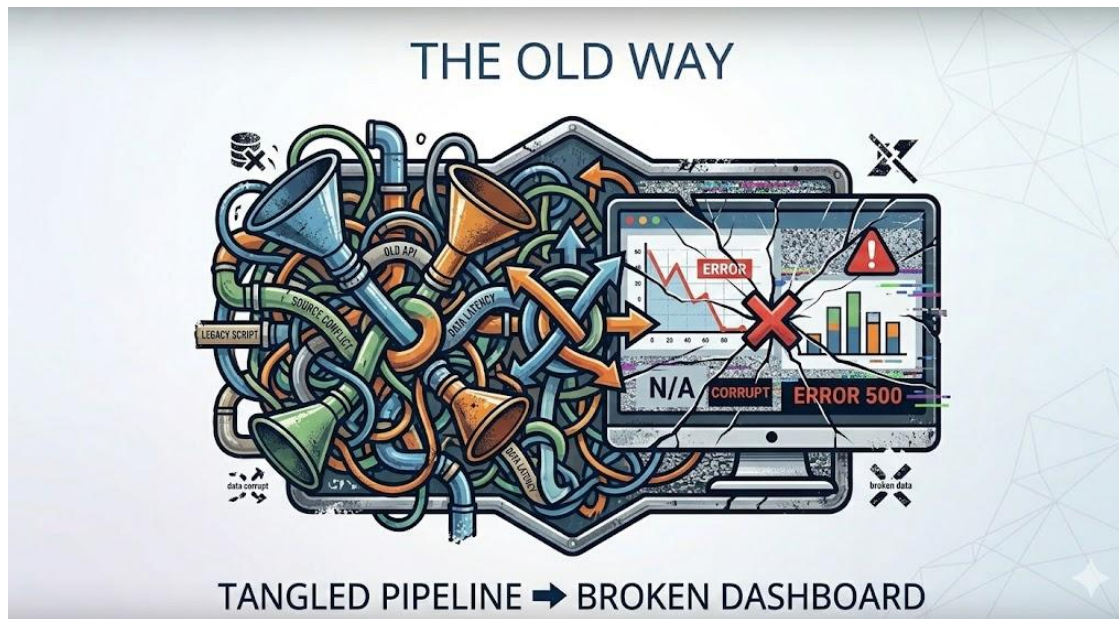
The Impact of Automation: Breaking the Silos



- 🧪 **Continuous Automated Testing**
 - QA in parallel; immediate failure detection.
- 🧠 **Zero Context Switching**
 - Developers fix fresh code instantly.
- 📊 **Proactive Monitoring**
 - Track CPU, memory, and logs in real-time.
- 🤖 **Automated Remediation**
 - Auto-rollbacks & context-rich feedback.
- 🏆 **The Ultimate Outcome**
 - Happy Teams (No blame game) + Happy Customers (Faster, stable releases).

Why DevOps Matters for Data Engineering?

The Challenge

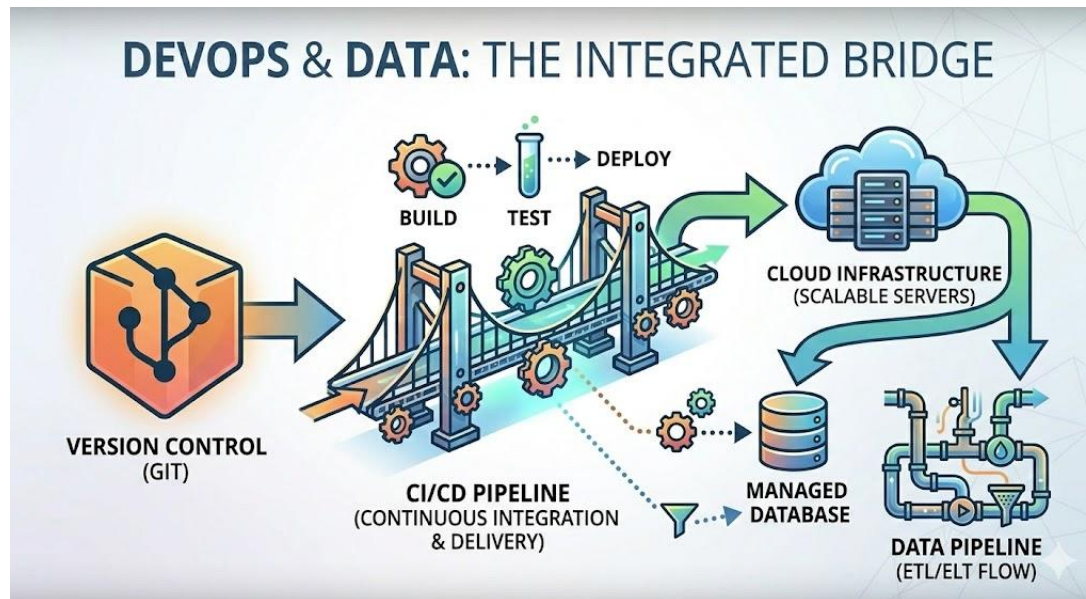


-  **Data Pipelines are Software**
 - They are no longer just simple SQL scripts.
-  **Massive Complexity**
 - Real-time streams, Big Data, multiple sources.
-  **The Data Silos**
 - Data Engineers vs. Data Scientists vs. Ops.
-  **The Cost of Failure**
 - Broken pipelines = Bad business decisions.
-  **Software Engineering vs Data Engineering**
 - Software = static code
 - vs
 - Data = static code + mutable data

The logo for xOps, featuring the text "xOps" in a white, bold, sans-serif font. The background is a dark blue gradient with a large, semi-transparent circle and a vertical line intersecting it.

xOps

Applying DevOps to Data (**DataOps**)



-  **Version Control Everything**
 - Code, database schemas, and queries.
-  **CI/CD for Data**
 - Automated testing for code *and* data quality.
-  **Infrastructure as Code (IaC)**
 - Automated provisioning of data environments (e.g., Spark clusters, Data Warehouses).
-  **Isolated Environments**
 - Dev, Test, and Prod for data pipelines.

The Ultimate Outcome: Trustable Data



-  **Data Reliability**
 - Catch anomalies before they reach the dashboard.
-  **Faster Time-to-Insight**
 - Rapid, safe deployment of new data models.
-  **Cross-Functional Collaboration**
 - Aligned goals across Data, Dev, and Ops.
-  **Business Trust**
 - Stakeholders trust the numbers they see.

Created DataOps to orchestrate and monitor **thousands of daily jobs**

Airbnb

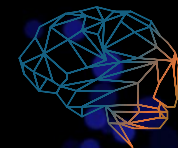
Airflow: a workflow management platform" (May 2015)

Processes over **100 billion events daily** to generate "Discover Weekly"

Spotify

The Evolution of Data Infrastructure at Spotify





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